

Individualized ECG waveform forecasting enables testable predictions of longitudinal morphology

Tao Wang¹, Zeming Huang¹, Yiquan Wang¹, Ning Liu¹,
Zhicheng Hu², Lei Wang³, Gong Su^{3*}, Zhipei Huang^{1*},
Ming Yin^{4*}, Fei Qin¹

¹ School of Electronic, Electrical and Communication Engineering,
University of Chinese Academy of Sciences, Beijing, 101408, China .

² Cardiac Arrhythmia Center, Fuwai Hospital, National Center for
Cardiovascular Diseases, Chinese Academy of Medical Sciences and
Peking Union Medical College, Beijing, 100037, China .

³ Cardiac Department, Aerospace Central Hospital, Beijing, 100049,
China .

⁴ The Second Medical Center and National Clinical Research Center for
Geriatric Diseases, Chinese PLA General Hospital, Beijing, 100853,
China .

*Corresponding author(s). E-mail(s): sugong@gt.cn;
zhphuang@ucas.ac.cn; mingyin@plagh.org;

Contributing authors: wangtao2025@ucas.ac.cn;
huangzeming24@mails.ucas.ac.cn; liuning251@mails.ucas.ac.cn;
mengxiangao17@mails.ucas.ac.cn; huzhicheng@fuwaihospital.org;
fromwanglei@126.com; fqin1982@ucas.ac.cn;

Abstract

Cardiovascular disease develops over years, yet electrocardiographic assessment remains snapshot-based. AI-ECG models compress distributed waveform information into ECG age, disease-risk scores, or diagnostic probabilities. However, such summaries and population-level age patterns do not describe how morphology changes from an individual’s baseline. We developed a generative AI-ECG framework that translates cross-sectional age-associated structure into source-conditioned future waveforms, rendering latent information as inspectable morphology. Forecasts agreed with within-person follow-up morphology. Among

baseline-negative MIMIC records with subsequently documented disease, forecast and observed follow-up ECGs had nearly identical positive rates (71.7% vs 71.8%) and 83.2% overlap in positive fixed-readout classifications, with similar correspondence in the private cohort. Similar baseline readout scores nevertheless corresponded to divergent forecast trajectories. Forecast-derived tiers showed graded outcome proportions in both cohorts. These findings support AI-based waveform forecasting as a tool for expressing future-associated ECG information as inspectable, testable morphology.

Keywords: electrocardiography, artificial intelligence, waveform forecasting, longitudinal modelling, generative modelling

1 Introduction

Clinical ECGs provide a widely available readout of cardiac electrical activation, conduction, and recovery [1]. In routine care, ECG interpretation is primarily centred on the current recording, providing an effective assessment of the present electrophysiological state while offering limited information about how that state changes over years. This distinction matters because cardiovascular disease evolves through cumulative cardiac remodelling [2], and conventional ECG measures have been associated with long-term cardiovascular outcomes [3–6]. Characterizing how ECG morphology

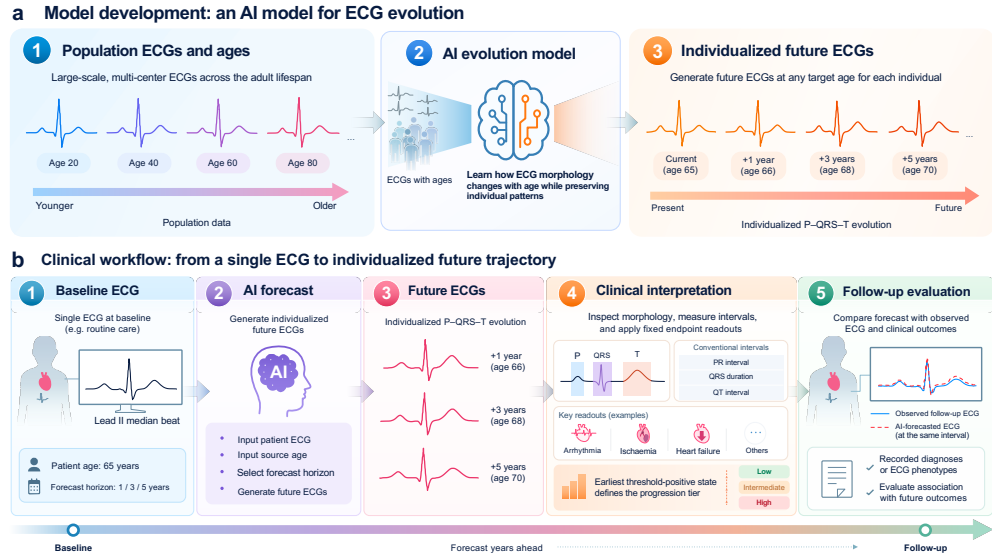


Fig. 1 Conceptual overview of individualized ECG waveform forecasting. **a**, Population ECGs are used to learn age-associated morphology. **b**, A baseline Lead II median beat and requested future age generate an individualized ECG trajectory, which is evaluated against subsequent within-person ECGs and clinical outcomes.

evolves within an individual may therefore reveal information that is not apparent from a single recording.

Chronological age provides a natural axis for studying longitudinal changes in ECG morphology. Age-related changes in ECG intervals and waveform morphology are well documented [7]. AI-ECG models further show that this age-related information is distributed across the waveform and can be summarized as ECG age [8, 9]. An ECG age exceeding chronological age by more than 8 years has been associated with increased mortality risk, including among ECGs considered clinically normal [10]. ECG age, however, remains a scalar summary of distributed waveform information. More generally, conventional AI-ECG models compress the current recording into compact age, risk, or diagnostic estimates. Such summaries can stratify individuals by risk, but similar values need not imply the same future waveform. They also do not specify which electrophysiological features may emerge or which waveform components may change, nor do they provide a physiological object that can be tested directly against the next ECG.

Recent advances have begun to express more of this information at the waveform level. Self-supervised models learn ECG representations directly from the signal [11, 12], while explainability and generative approaches can map predictive information to interpretable morphology [13, 14]. Conditional models can further synthesize ECGs under demographic, diagnostic, or textual conditions [15–18]. Together, these developments show that clinically relevant ECG information can be represented, interpreted, and manipulated directly at the waveform level. This creates the basis for moving beyond characterization of the current recording toward modelling how waveform morphology evolves within an individual over time. The resulting task is not simply to synthesize an ECG at a target age, but to translate population-level age-associated structure into a source-conditioned longitudinal trajectory anchored to the individual’s baseline.

This longitudinal evolution problem differs fundamentally from conventional time-series forecasting. Multi-year ECG evolution is observed only through sparse and irregular clinical recordings and reflects slow electrophysiological remodelling rather than short-term signal continuation. Three linked challenges therefore arise. First, stable person-specific morphology is intertwined with evolving physiological variation, so forecasting must express change without replacing the individual’s baseline structure. Repeated-recording studies demonstrate the persistence of individual ECG morphology over years [19, 20], while longitudinal representation learning has sought to distinguish stable individual characteristics from evolving physiological state [21, 22]. Second, forecasts requested at neighbouring ages should form a coherent morphological trajectory rather than unrelated age-conditioned samples. Third, sparse same-person recordings leave most intermediate states unobserved and limit direct supervision with future waveform targets. Population age differences must therefore be translated into source-conditioned change and tested against longitudinal observations, rather than treated as within-person progression by assumption.

We present an AI-based waveform forecasting tool that expresses future-associated ECG information as inspectable morphology. An overview of the model development

and longitudinal evaluation workflow is shown in Fig. 1. The framework learns population age-associated structure while retaining characteristics of the individual’s baseline ECG. An age-indexed Gaussian memory organizes the learned structure, and source-conditioned adaptive modulation determines its expression at a requested future age. The resulting Lead II median beat can be compared directly with a later recording and examined using conventional measurements and fixed readouts.

We evaluate this framework through a sequence of representation, waveform, and clinical follow-up analyses. We first characterize the learned age-associated representation and compare generated waveforms with within-person follow-up ECGs and comparator forecasts. We then examine fixed-readout agreement and use coarsened exact matching to compare forecasted and observed futures among records with comparable baseline readout scores. Finally, we assess associations between forecast-derived tiers and recorded outcomes, characterize regional changes along the generated trajectories, and evaluate retention of baseline disease discrimination.

2 Results

The development pool comprised 1,041,620 ECGs from 387,180 participants, partitioned at the participant level. The waveform forecasting model was trained using cross-sectional age-labelled ECGs without observed future waveform targets. Longitudinal evaluation drew on a held-out MIMIC subset and two private data resources. The waveform-validation archive contained 124,373 ECGs, with 72,381 identifiable patients; the clinical subset comprised 13,553 ECGs from 959 participants for fixed-readout and outcome analyses. Source (G_0) denotes the baseline ECG; G_1 , G_3 , and G_5 denote waveforms generated at 1, 3, and 5 years after baseline, respectively.

2.1 Age-associated variation is concentrated in the evolution representation

The evolution and invariant representations were designed to capture age-associated variation and source-preserving morphology, respectively, as described in Methods 4.4. Along the learned age-associated trajectory, the source-balanced centroid showed greater displacement in the evolution space, whereas the invariant representation remained comparatively stable (Figure 2a).

Matched perturbations produced distinct waveform changes in the two representations. Movement in the evolution space produced structured alterations across P–QRS–T morphology, whereas matched movement in the invariant space produced broader changes in source-specific waveform shape (Figure 2b). These analyses support the intended separation of age-associated variation from source-preserving morphology within the learned representation.

2.2 Forecasted waveforms show agreement with within-person follow-up morphology

We quantified waveform agreement between forecasted ECGs and subsequent recordings from the same individuals in the private waveform-validation archive. Waveform

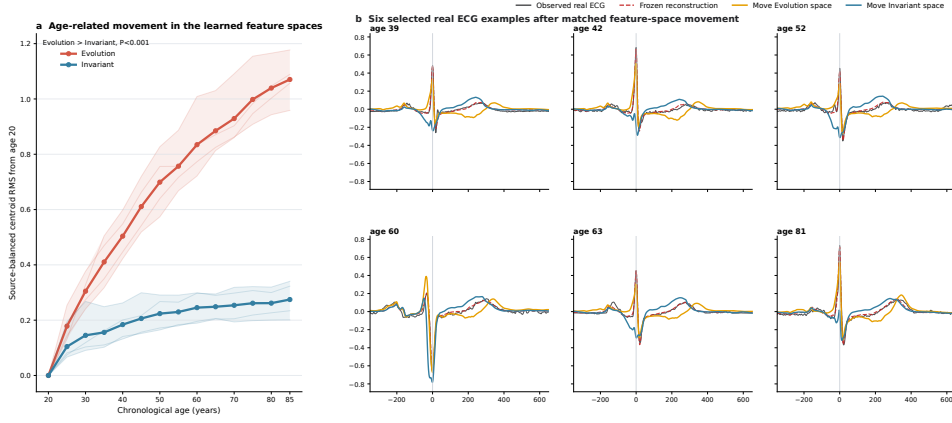


Fig. 2 Age-associated variation concentrates in the evolution representation. **a**, Source-balanced centroid root-mean-square displacement from the age-20 reference across chronological age in the evolution and invariant spaces. Lines and shaded regions summarize the central trajectory and its variation across source-balanced estimates. **b**, Six selected ECGs after matched displacement in either representation. Black lines denote observed ECGs, dashed red lines denote frozen reconstructions, orange lines denote movement in the evolution space, and blue lines denote movement in the invariant space.

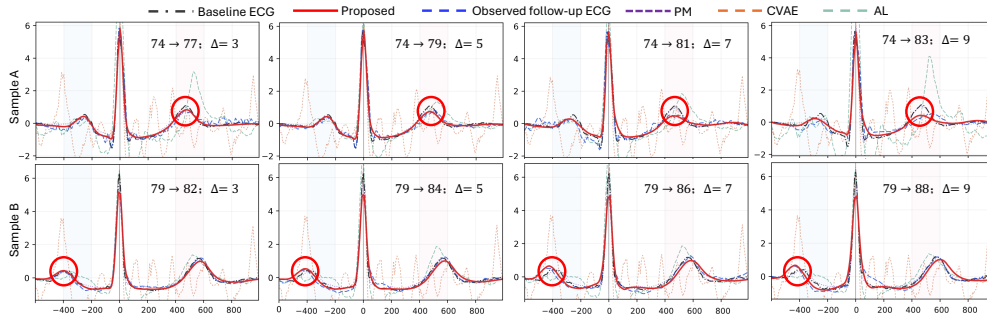


Fig. 3 Examples of forecasted and observed within-person follow-up morphology. Sample A shows forecasts from a baseline age of 74 years and Sample B from a baseline age of 79 years across increasing forecast horizons. All forecasting methods were evaluated on identical source-target pairs using the same preprocessing and temporal alignment, and none used observed longitudinal target waveforms for forecasting-model training. CVAE denotes conditional variational autoencoder, AL denotes AgeLatent, and PM denotes PointMemory; comparator definitions are provided in Methods 4.7.

agreement was compared with CVAE, AgeLatent, and PointMemory (Methods 4.7). The paired examples show preservation of source-specific morphology alongside changes toward the corresponding follow-up ECGs (Figure 3). In Sample A, source-specific QRS morphology remained stable while lower-amplitude P- and T-wave features shifted toward the observed target. In Sample B, the prolonged PR interval in the source ECG was preserved across forecast horizons. In these examples, CVAE

Method	NMSE $\times 10^3 \downarrow$	P/T NMSE $\times 10^3 \downarrow$	P/T dNMSE $\times 10^5 \downarrow$	Peak position error (samples) $\downarrow$
Proposed model	1.411$\pm$0.819	0.562$\pm$0.013	4.917$\pm$0.178	3.372$\pm$0.111
PM	1.482 $\pm$ 0.844	0.651 $\pm$ 0.015	5.604 $\pm$ 0.183	3.756 $\pm$ 0.121
AL	6.311 $\pm$ 2.707	5.354 $\pm$ 0.166	23.16 $\pm$ 0.761	6.188 $\pm$ 0.156
CVAE	15.681 $\pm$ 3.703	24.77 $\pm$ 0.249	286.9 $\pm$ 2.794	20.24 $\pm$ 0.254

Table 1 Waveform agreement with observed follow-up ECGs across forecasting models (mean $\pm$ SD). All methods were evaluated on identical source–target pairs using the same preprocessing and waveform metrics, and none used observed longitudinal target waveforms for forecasting-model training. CVAE denotes conditional variational autoencoder, AL denotes AgeLatent, and PM denotes PointMemory; comparator definitions are provided in Methods 4.7.

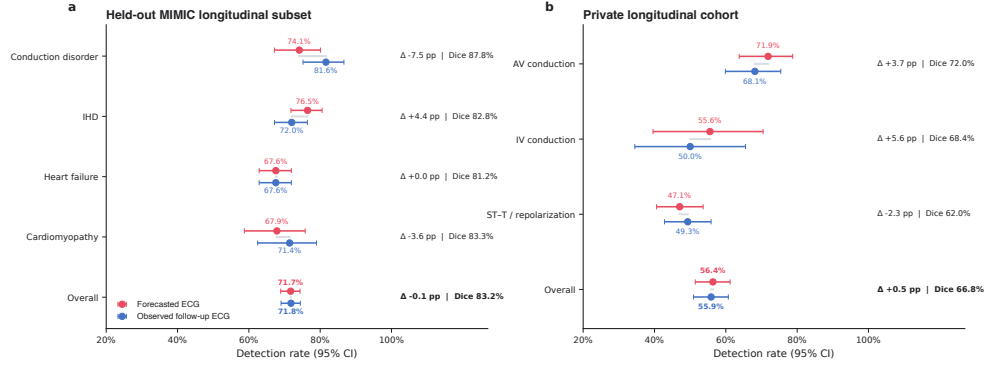


Fig. 4 Fixed-readout correspondence between forecasted and observed follow-up ECGs. Points show detection rates with 95% confidence intervals for forecasted ECGs and observed follow-up ECGs. Analyses include baseline-negative endpoint opportunities with a subsequently documented clinical diagnosis or ECG phenotype. The waveform forecasting model was applied to the private cohort without adaptation. Annotations report forecast-minus-observed differences in percentage points and Dice coefficients for binary-label agreement.

produced oscillatory artefacts, AL showed larger local waveform deviations, and PM produced smaller but persistent mismatches with the observed targets.

The proposed model achieved the lowest error across all reported waveform-level measures (Table 1). Full-waveform NMSE was 1.411×10^{-3} . Relative to PM, P/T-window NMSE was reduced by 13.7% and P/T derivative-NMSE by 12.3%, together with a lower peak-position error.

2.3 Forecasted and observed follow-up ECGs show fixed-readout correspondence

We next used the fixed waveform-only ECG readouts described in Methods 4.5 to assess endpoint-associated features in forecasted and observed follow-up ECGs. Analyses included endpoint opportunities with a negative recorded clinical diagnosis at baseline, followed by a documented positive endpoint during follow-up. Each endpoint was evaluated with the same readout and operating threshold for both waveform types.

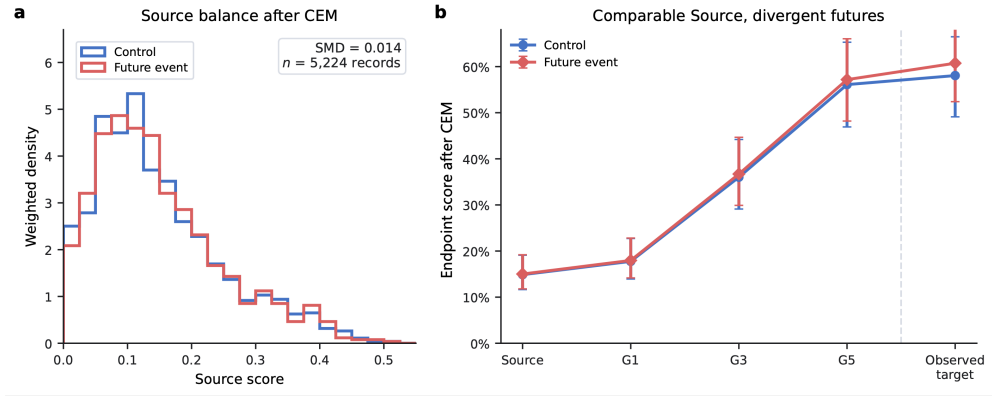


Fig. 5 Comparable Source states diverge along forecasted and observed future trajectories in MIMIC. **a**, Source-score distributions for future-event and control records after coarsened exact matching. Matching was performed within endpoint and 0.05-wide Source-score intervals, retaining 5,224 records and achieving a Source-score standardized mean difference of 0.014. **b**, CEM-adjusted mean endpoint scores at Source, G_1 , G_3 , G_5 , and the observed follow-up target. Points and error bars denote means and 95% confidence intervals from 5,000 CEM-stratum bootstrap resamples.

In the held-out MIMIC longitudinal subset, the classifier-positive rate was 71.7% for forecasted ECGs and 71.8% for observed follow-up ECGs overall (Figure 4). The classifier-positive rates were identical for heart failure (67.6% versus 67.6%) and differed by 7.5 percentage points for conduction disorder (74.1% versus 81.6%). Binary-label agreement between forecasted and observed ECGs yielded an overall Dice coefficient of 83.2%, with endpoint-specific values ranging from 81.2% to 87.8%.

The waveform forecasting model was applied to the private longitudinal cohort without adaptation. The overall classifier-positive rate was 56.4% for forecasted ECGs and 55.9% for observed follow-up ECGs, with a Dice coefficient of 66.8%. Endpoint-level rates were 71.9% versus 68.1% for atrioventricular conduction abnormalities, 55.6% versus 50.0% for intraventricular conduction abnormalities, and 47.1% versus 49.3% for ST-T/repolarization abnormalities. Endpoint-specific Dice coefficients ranged from 62.0% to 72.0%.

Using coarsened exact matching as described in Methods 4.6, we evaluated whether future-event and control records with comparable baseline readout scores showed different forecasted and observed follow-up scores. Coarsened exact matching within endpoint and 0.05-wide Source-score intervals retained 5,224 of 5,225 eligible MIMIC records and closely balanced the Source-score distributions of future-event and control records (standardized mean difference, 0.014; Figure 5a). The event-control difference was 0.14 percentage points at Source and expanded along the forecasted trajectory to 0.21 percentage points at G_1 (95% CI, 0.04–0.34), 0.67 at G_3 (0.34–0.95), and 1.08 at G_5 (0.53–1.58). The corresponding difference in the observed follow-up ECG reached 2.68 percentage points (95% CI, 1.48–3.77; Figure 5b). Despite closely balanced baseline readout scores, event-control score differences increased across the generated forecast states and were larger in the observed follow-up ECGs.

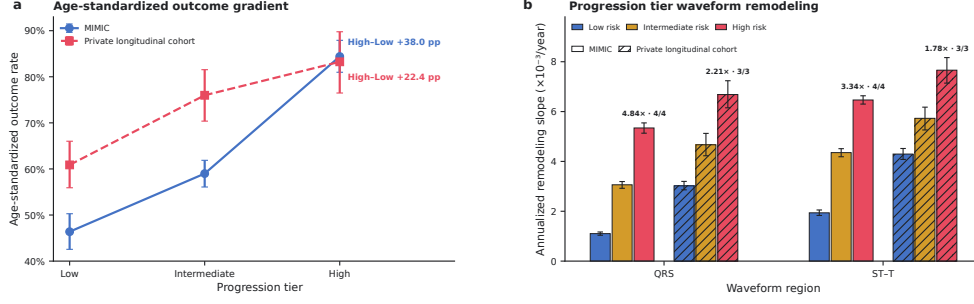


Fig. 6 Recorded outcomes and generated waveform changes across forecast-derived tiers. **a**, Age-standardized recorded follow-up outcome proportions in the Low, Intermediate, and High progression tiers. **b**, Annualized QRS and ST-T changes calculated along the generated trajectories across the same tiers. Annotations show High-to-Low slope ratios and the number of endpoints showing consistent ordering. Error bars denote 95% bootstrap confidence intervals.

2.4 Forecast-derived tiers are associated with recorded follow-up outcomes

We next used the predefined forecast-derived progression tiers to examine their association with recorded follow-up outcomes. Patients were grouped into forecast-derived progression tiers: Low when no eligible endpoint crossed its threshold at G_1 , G_3 or G_5 ; Intermediate when the first crossing occurred at G_5 ; and High when the first crossing occurred at G_1 or G_3 .

Age-standardized recorded follow-up outcome proportions increased monotonically across progression tiers in both longitudinal cohorts (Figure 6a). The High-minus-Low difference was 38.0 percentage points in MIMIC and 22.4 percentage points in the private cohort. This ordering shows a graded association between forecast-derived tiers and recorded follow-up outcomes across cohorts with different clinical compositions and endpoint definitions.

Regional changes along the generated trajectories also increased across tiers (Figure 6b). In MIMIC, the High-to-Low ratio of annualized waveform change was 4.84 for QRS and 3.34 for ST-T. The corresponding ratios were 2.21 and 1.78 in the private cohort. The ordering was consistent across all four MIMIC endpoints and all three private-cohort endpoints.

Selected forecast trajectories illustrated the morphology underlying this tier ordering (Figure 7). Examples in the Low tier showed limited separation across forecast horizons, whereas those in the Intermediate and High tiers showed progressively greater deviation from baseline, particularly in post-QRS morphology. The dominant source-specific waveform structure remained recognizable across forecast states.

2.5 Endpoint-associated forecast changes are concentrated in QRS and ST-T regions

We characterized the regional distribution of generated waveform changes associated with subsequently recorded endpoints. Event-control differences in annualized

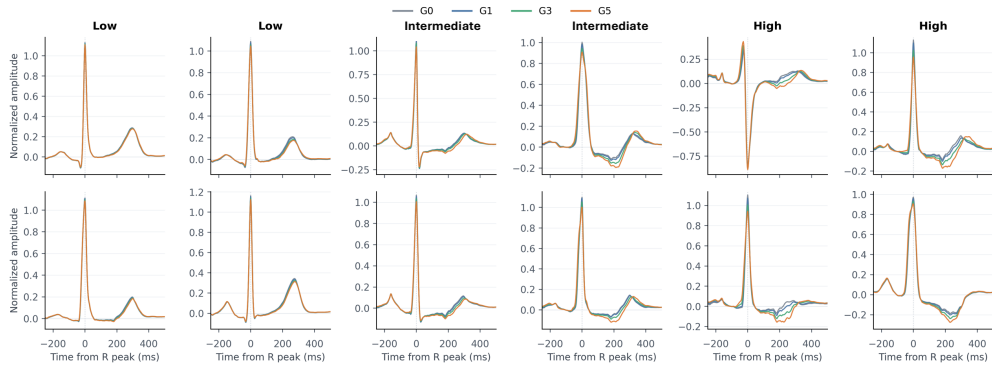


Fig. 7 Selected examples from the Low, Intermediate, and High progression tiers are shown across forecast states. G_0 denotes the baseline ECG, and G_1 , G_3 , and G_5 denote the generated 1-, 3-, and 5-year states, respectively.

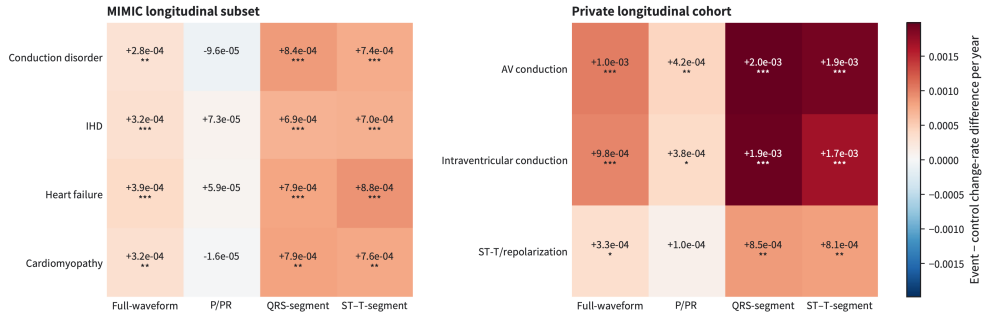


Fig. 8 Regional distribution of endpoint-associated changes in generated waveforms. Cells show event-minus-control differences in annualized waveform-change rate for the full waveform and P/PR, QRS, and ST-T regions.

waveform-change rate were calculated for the full waveform and the P/PR, QRS, and ST-T regions.

Event-control differences in annualized generated waveform change were largest in the QRS and ST-T regions (Figure 8). Across MIMIC endpoints, these differences ranged from approximately 6.9×10^{-4} to 8.4×10^{-4} per year in QRS and from 7.0×10^{-4} to 8.8×10^{-4} per year in ST-T. P/PR differences were smaller and less consistent in direction. The private cohort showed the same regional pattern, with the largest QRS and ST-T differences observed for atrioventricular and intraventricular conduction abnormalities. Horizon-specific regional trajectories are shown in Supplementary Fig. 3.

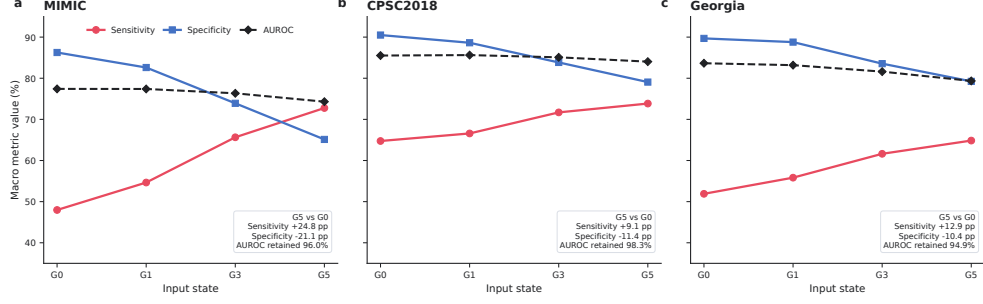


Fig. 9 Retention of baseline disease discrimination across generated states. Macro sensitivity, specificity, and AUROC are shown at baseline (G_0) and at generated 1-, 3-, and 5-year states (G_1 , G_3 , and G_5). All states are evaluated with the same frozen classifiers and fixed operating thresholds.

2.6 Baseline disease discrimination is largely retained across forecast states

Finally, we assessed whether longitudinal forecasting altered disease-relevant information already present in the source ECG. The same frozen classifiers and fixed thresholds were applied to baseline and generated 1-, 3-, and 5-year states in MIMIC, CPSC2018, and Georgia.

At five years, macro AUROC retained 96.0% of its baseline value in MIMIC, 98.3% in CPSC2018, and 94.9% in Georgia (Figure 9). At the fixed operating thresholds, sensitivity increased and specificity decreased with forecast horizon, indicating a shift toward classifier-positive states. Endpoint-level operating metrics are shown in Supplementary Fig. 4.

3 Discussion

We learned age-associated ECG structure from cross-sectional recordings and used it to generate individualized Lead II forecasts that were evaluated against subsequent ECGs from the same person. Agreement at waveform and fixed-readout levels, together with separation after CEM and associations between forecast-derived tiers and recorded outcomes, supports the feasibility of waveform forecasting. These findings connect age-associated patterns learned across individuals to observable longitudinal variation within individuals.

ECG-age studies have linked waveform-derived age estimates to cardiovascular outcomes [10, 23]. Here, age-associated structure is expressed as Lead II median-beat morphology, placing a prediction in the same domain as the subsequent recording. The tool produces waveforms that can be inspected across P-QRS-T morphology and examined using conventional measurements and fixed readouts. The CEM analysis further distinguishes trajectory forecasting from static risk scoring. Records with closely comparable Source scores subsequently separated along the forecasted trajectory, and this separation was followed by a larger difference in the observed follow-up

ECG. A scalar Source score summarizes one aspect of the baseline ECG but does not specify the subsequent waveform trajectory.

The framework combines population age-associated variation with individual ECG structure. Separating the evolution and source-preserving representations allows morphology to change along the modelled age direction while retaining characteristics of the baseline ECG, building on longitudinal representation approaches that separate individual characteristics from evolving state [21, 22]. The greater age-associated displacement of the evolution representation and the distinct waveform effects of matched perturbations support this functional separation. The Gaussian memory organizes population age structure, while source-conditioned adaptive intensity controls its expression from each baseline ECG. Evaluation against held-out follow-up recordings connects this population-based construction to within-person morphology.

Waveform comparison tests both preservation of individual structure and agreement with local changes in subsequent recordings. Full-waveform error summarizes overall morphology, whereas P/T-window error, derivative-based error, and peak-position error examine lower-amplitude features, local shape, and temporal alignment. Together, these measurements relate forecast agreement to specific waveform properties and locations. The lower P/T errors relative to PointMemory show that the improvement extends to lower-amplitude atrial and repolarization features. The paired examples further illustrate these differences: the proposed forecasts retained source-specific QRS morphology while showing smaller local deviations than CVAE and AgeLatent.

Fixed readouts connect the waveform comparison to features associated with subsequent clinical diagnoses or ECG phenotypes. Applying the same readout and threshold to generated and observed follow-up ECGs places both on a common measurement scale. Positive-call proportions describe how often the corresponding features are detected, whereas paired Dice overlap describes whether they occur in the same endpoint opportunities. Their correspondence in MIMIC and the private cohort links the generated morphology to endpoint-associated patterns in the observed follow-up recordings.

Forecast-derived tiers were associated with graded recorded outcome proportions in both longitudinal cohorts, linking the timing of threshold crossing in the forecasts to subsequent clinical follow-up. Regional analyses showed how those tiers were expressed within the generated waveforms, with larger changes in QRS and ST-T regions. This waveform description extends work that uses explainability and generative methods to express predictive ECG information as morphology [13, 14, 23, 24]. Baseline disease discrimination remained largely preserved across forecast states, indicating that the generated changes retained information relevant to the baseline labels. Increasing sensitivity and decreasing specificity indicate a shift toward classifier-positive states at fixed thresholds.

Age provides a common coordinate for requesting future waveform states, while the shared age-associated direction and bounded adaptive intensity organize change from each baseline ECG. This construction produces a sequence whose regional morphology can be examined across requested ages, as illustrated by the QRS and ST-T analyses.

Extending the representation to multilead and multibeat ECGs would bring spatial and beat-to-beat variation into the same forecast–observation comparison.

Individualized ECG waveform forecasting makes a future morphological hypothesis available for direct inspection and longitudinal testing. This approach connects baseline ECG information to longitudinal cardiac phenotypes through waveforms that can be evaluated against subsequent recordings.

4 Methods

4.1 Study objective and evaluation design

This retrospective study evaluated the feasibility of generating source-conditioned future ECG morphology from cross-sectional age-associated structure. The waveform forecasting model was developed without observed future waveform targets. Evaluation comprised comparison with within-person follow-up ECGs, fixed-readout analyses, associations with recorded follow-up outcomes, and assessment of baseline disease-discrimination retention. The waveform representation was an R-aligned Lead II median beat.

4.2 Datasets and study cohorts

Model development used CODE-15%, MIMIC-IV-ECG, PTB-XL, and Shaoxing [25–28]. After waveform quality control, the development pool comprised 1,041,620 ECGs from 387,180 participants aged 18–95 years and was partitioned at the participant level into approximately 80% training, 10% validation, and 10% test sets. Longitudinal clinical evaluation used a held-out MIMIC-IV-ECG subset comprising 5,269 endpoint-records from 2,454 participants; none contributed to forecasting-model or clinical-readout development, calibration, or threshold selection.

Private longitudinal data were obtained from Fuwai Hospital and Aerospace Central Hospital. A curated clinical subset comprised 13,553 ECGs from 959 participants and was partitioned at the participant level into training, validation, and locked-test sets; only the locked-test partition was used for disease-associated longitudinal evaluation. The public-data-trained forecasting model was applied without private-cohort adaptation. A separate waveform-validation archive comprised 124,373 twelve-lead ECGs from 72,381 identifiable patients, including 24,147 patients with repeated examinations and follow-up of up to 4,122 days, and was used for direct comparison of generated waveforms with subsequent same-person ECGs. Data use and retrospective analysis were approved by the relevant institutional ethics committees, with informed consent waived.

CPSC2018 and Georgia were additionally used as independent held-out datasets for disease-discrimination retention [29, 30], with final test sets of 635 and 886 ECGs, respectively.

4.3 ECG preprocessing and waveform representation

Lead II signals were calibrated to millivolts and resampled to 500 Hz when necessary. After record-level quality control and baseline correction, R-aligned beats were

extracted, screened for morphology consistency, and combined by pointwise median without temporal warping or nonlinear alignment. The median beat was low-pass and notch filtered, resampled to 400 Hz, and represented as a 512-sample waveform with the R peak fixed at zero-based index 208. No amplitude normalization or clipping was applied, and recordings failing signal, beat-count, morphology, or final-output quality checks were excluded rather than imputed.

4.4 Model development

The forecasting backbone was a one-dimensional residual variational encoder–decoder operating on the R-aligned Lead II median beat. Each ECG was represented by an 8-dimensional evolution representation, $\mathbf{z}_{\text{evo}}$, and a 40-dimensional invariant, source-preserving representation, $\mathbf{z}_{\text{inv}}$. The decoder reconstructed the waveform from both components. A supervised age head encouraged age-associated ordering in the evolution representation, whereas an adversarial age head coupled through gradient reversal reduced age information in the invariant representation. The principal model components are summarized in Supplementary Table 1, and the latent separation is illustrated in Supplementary Fig. 1.

Cross-sectional age-labelled ECGs were used to organize the evolution representation with an age-indexed Gaussian memory. Overlapping Gaussian neighborhoods around ordered age anchors defined smoothly varying age-conditioned latent means and variances. The evolution posterior was aligned with the corresponding age-conditioned Gaussian state using KL divergence, while second-order smoothness regularization discouraged abrupt changes between adjacent age anchors. The ordered memory states were then used to derive a shared population age direction, $\mathbf{v}_{\text{global}}$. The Gaussian memory structure is illustrated in Supplementary Fig. 2.

After cross-sectional training, the encoder–decoder backbone, age head, and Gaussian memory were frozen. A source-conditioned Adaptive-global intensity network was then trained using cross-sectional source ECGs with sampled forward target ages, without observed longitudinal target waveforms. The network combined the source evolution representation, evolution-dispersion features, invariant representation, and source–target age descriptors to determine a bounded positive intensity, $I_{s,t}$. For source age a_s and requested target age a_t , the target latent representations were

$$\begin{aligned}\mathbf{z}_{\text{evo},t} &= \mathbf{z}_{\text{evo},s} + \mathbf{v}_{\text{global}}(a_t - a_s)I_{s,t}, \\ \mathbf{z}_{\text{inv},t} &= \mathbf{z}_{\text{inv},s}.\end{aligned}\tag{1}$$

Thus, the population direction specifies the direction of age-associated change, the source–target age difference specifies the forecast horizon, and the adaptive intensity controls how strongly that direction is expressed from each source ECG. At inference, the model required only the source waveform, source age, and requested target age. Observed longitudinal ECGs were reserved for held-out waveform and clinical evaluation.

4.5 Clinical endpoints and fixed ECG readouts

Endpoint opportunities required a negative source label, future-positive status was assigned at the first documented endpoint-positive event within follow-up, whereas insufficient follow-up or uninterpretable labels were masked (endpoints were derived from MIMIC-IV-ECG-Ext [31]). Endpoint-associated waveform features were evaluated using a fixed single-waveform AI-ECG readout that received only the 512-sample median beat, without age, forecast horizon, demographic variables, or additional waveforms. The readout used a one-dimensional residual squeeze-and-excitation network with a four-output classification head for the MIMIC endpoints.

Private-cohort endpoints were ECG phenotypes comprising atrioventricular conduction abnormality, intraventricular conduction abnormality, and ST-T/repolarization abnormality, mapped from the prespecified SNOMED-style coding contract. The same waveform-only readout architecture was used with a three-output classification head for these phenotypes. The locked-test partition was excluded from readout training, calibration, and threshold selection, and the frozen readout was applied independently to Source, G_1 , G_3 , G_5 , and observed follow-up waveforms.

4.6 Evaluation and statistical analysis

Direct waveform validation compared forecasts generated at the observed source-target age intervals with the corresponding follow-up ECGs. Standardized 1-, 3-, and 5-year forecast states (G_1 , G_3 , and G_5) were additionally used for multi-horizon analyses. Waveform agreement was assessed using full-waveform NMSE, P/T-window NMSE, P/T derivative NMSE, and peak-position error, and was compared across the proposed model, CVAE, AgeLatent, and PointMemory. All forecasting models were evaluated on the same source-target pairs using identical preprocessing and metrics.

For eligible endpoint opportunities, the same frozen readout and validation-selected threshold were applied to Source, generated, and observed follow-up ECGs. Detection proportions were reported with Wilson 95% confidence intervals, and paired forecast-observation agreement was quantified using the Dice coefficient.

Coarsened exact matching compared future-event and control records within endpoint and 0.05-wide Source-score intervals. Source-score balance was assessed using the standardized mean difference. Event-control differences at Source, G_1 , G_3 , G_5 , and observed follow-up were estimated after CEM, with 95% confidence intervals obtained from 5,000 CEM-stratum bootstrap resamples. Progression tiers were defined by the earliest threshold crossing across G_1 , G_3 , and G_5 , and recorded outcome proportions were age-standardized with 95% confidence intervals from 2,000 patient-level bootstrap resamples.

Regional waveform change was quantified within the complete waveform and predefined P/PR, QRS, and ST-T regions. Cross-sectional disease discrimination in MIMIC, CPSC2018, and Georgia was assessed using AUROC, sensitivity, specificity, and F1 at fixed validation-selected thresholds, with endpoint values macro-averaged within each dataset.

4.7 Baseline implementations

The comparator models were designed to test alternative ways of translating cross-sectional age information into target-age ECG waveforms. Observed longitudinal target waveforms were not used to train the comparator forecasting operators; longitudinal source–target ECGs were used only for final waveform evaluation.

Conditional variational autoencoder.

The conditional variational autoencoder (CVAE) directly conditioned the latent generative process on chronological age [32]. At inference, the source ECG was encoded at its observed age and decoded using the requested target age. The model therefore tested whether direct age conditioning was sufficient to generate a source-consistent target-age waveform without explicit separation of age-associated and source-preserving representations.

AgeLatent.

The AgeLatent comparator used an evolution–invariant latent decomposition but did not include the Gaussian age memory. The evolution component was encouraged to encode age-associated variation, whereas adversarial training reduced age information in the invariant component. Target-age forecasting was obtained by shifting the evolution representation according to an age-dependent latent transport while retaining the invariant representation.

PointMemory.

PointMemory augmented the disentangled backbone with an age-indexed deterministic memory. Unlike the proposed Gaussian memory, each age state was represented by a point estimate of the latent mean. Forecasting used the displacement between the queried source-age and target-age memory means to modify only the evolution representation while preserving the invariant representation.

All comparator implementations, optimization settings, and evaluation scripts are provided in the accompanying source-code repository.

Data availability

CODE-15%, MIMIC-IV-ECG, PTB-XL, Shaoxing, CPSC2018, and Georgia are available from their original public repositories under the corresponding access and data-use conditions. CODE-15% is available through Zenodo at <https://zenodo.org/records/4916206> [25]. MIMIC-IV-ECG and its extended ICD-label resource are available through PhysioNet at <https://physionet.org/content/mimic-iv-ecg/1.0/> and <https://physionet.org/content/mimic-iv-ecg-ext-icd-labels/1.0.1/>, respectively [26, 31, 33]. PTB-XL is available at <https://physionet.org/content/ptb-xl/1.0.3/> [27], and Shaoxing is available at <https://physionet.org/content/ecg-arrhythmia/1.0.0/> [28]. CPSC2018 and Georgia are distributed through the PhysioNet/Computing in Cardiology Challenge 2020 collection at <https://physionet.org/content/challenge-2020/1.0.2/> [29, 30].

The private longitudinal ECG cohort cannot be publicly released because the clinical waveform data are owned and governed by the collaborating hospitals and are subject to institutional restrictions on redistribution. Requests for access may be directed to the corresponding authors. Any data sharing will require approval from the data-owning institutions, the applicable research-ethics authorization, and an executed data-use agreement.

Code availability

The full preprocessing, model-development, forecasting, readout, and statistical-analysis implementations will be released with the accompanying source-code repository.

Author contributions

T.W.: Conceptualization, Methodology, Software, Formal analysis, Investigation, Visualization, Writing—original draft. Z.M.H., Y.W., and N.L.: Investigation, Data curation, Formal analysis, Validation. Z.C.H., G.S., L.W., and M.Y.: Resources, Data curation, Validation, Clinical interpretation, Conceptualization. Z.P.H. and F.Q.: Conceptualization, Methodology, Supervision, Writing—review and editing. All authors reviewed and approved the final manuscript.

Competing interests

The authors declare no competing interests.

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Supplementary Information

Network architecture

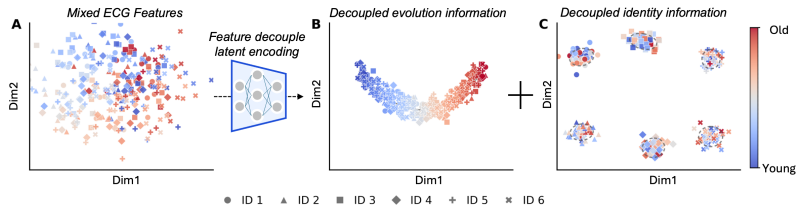
Component	Structure	Function
Input	Lead II, 1×512	R-aligned median beat
Encoder	1D residual variational encoder	Latent Gaussian representation
Evolution latent	8 dimensions	Age-associated variation
Invariant latent	40 dimensions	Source-preserving morphology
Age head	Supervised prediction head	Age ordering of evolution latent
Adversarial age head	Prediction head + GRL	Age suppression in invariant latent
Gaussian memory	Age-indexed Gaussian states	Population age structure
Adaptive intensity	$61 \rightarrow 128 \rightarrow 64 \rightarrow 1$	Source-conditioned movement magnitude
Decoder	1D residual decoder	Waveform reconstruction/generation

Supplementary Table 1 Architecture and functional roles of the principal components of the waveform forecasting model.

The model was organized around two complementary latent representations. The evolution component provided the space in which age-associated variation was organized and subsequently displaced during forecasting, whereas the invariant component retained source-specific morphology that was not directly modified by the progression operation. The Gaussian memory and Adaptive-global intensity module acted on this separation at different levels: the memory represented population age-associated structure, while the adaptive module controlled how strongly that structure was expressed from an individual source ECG.

This organization was intended to separate three functions that would otherwise be entangled in a single latent representation: preservation of the source waveform, representation of population age-associated variation, and modulation of future change from a specific baseline. The corresponding functional behavior of these components is examined in the latent-space and waveform analyses reported in the main text.

Latent separation

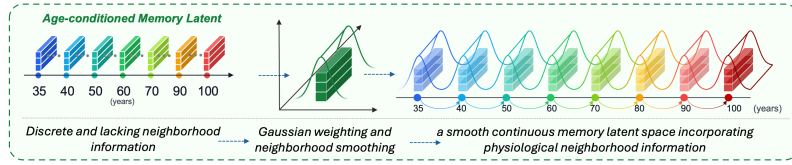


Supplementary Fig. 1 Schematic of latent separation. The evolution representation is encouraged to retain age-ordered variation, whereas the invariant representation preserves source-specific morphology with reduced age ordering.

The two latent components were designed to play complementary roles rather than to form mutually exclusive representations of ECG morphology. The evolution representation retained variation associated with chronological age, whereas the invariant representation retained information required to reconstruct the source waveform while reducing explicit age ordering.

This distinction is important for individualized forecasting. A future waveform should not be generated by replacing the source ECG with a population-average waveform at the requested age. Instead, age-associated change is expressed through the evolution representation while the invariant representation anchors the forecast to the source morphology. The main-text latent-space analysis provides empirical support for this intended separation by showing greater age-associated displacement in the evolution representation and distinct waveform effects under matched perturbations.

Age-indexed Gaussian memory and population age direction



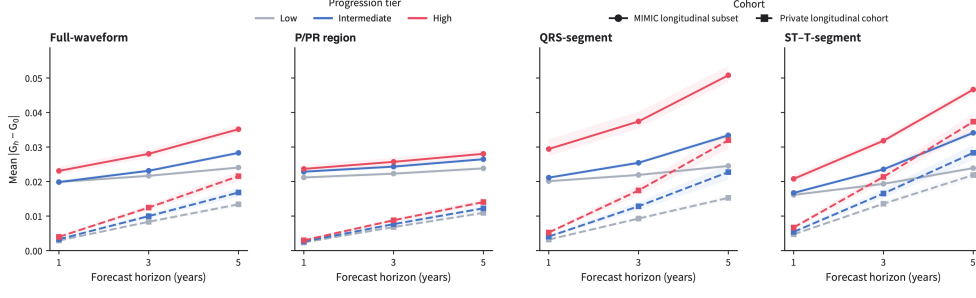
Supplementary Fig. 2 Evolution representations are organized by overlapping Gaussian age neighborhoods, with age-label sharing across nearby anchors and smoothness regularization of the memory geometry.

The Gaussian memory organizes cross-sectional evolution representations along chronological age without treating individual age groups as isolated states. Overlapping age neighborhoods allow nearby ages to contribute to adjacent memory states, while smoothness regularization discourages abrupt changes in the learned population trajectory.

The ordered memory therefore provides a continuous population reference from which a shared age-associated direction can be derived. During forecasting, this direction determines where the evolution representation moves, while the requested age interval determines how far the forecast progresses and the Adaptive-global intensity determines how strongly the shared direction is expressed from the source ECG. The memory itself does not use observed future ECGs and is learned entirely from cross-sectional age-labelled recordings.

Regional trajectories over forecast horizon

Regional waveform trajectories showed progressively greater separation across forecast horizons. The separation was most evident in the QRS and ST-T regions, whereas the P/PR region showed smaller and less consistent differences between progression tiers. This pattern was observed in both MIMIC and the private cohort despite their different endpoint definitions and clinical compositions.



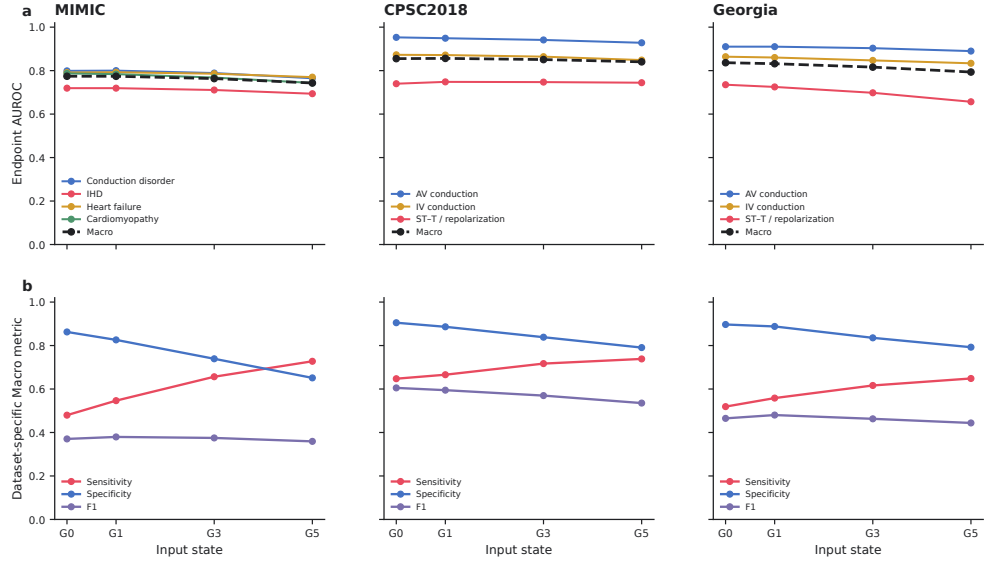
Supplementary Fig. 3 Mean absolute waveform change from baseline at 1-, 3-, and 5-year forecast states across the complete waveform and P/PR, QRS, and ST-T regions. Colours denote forecast-derived progression tiers; solid circles denote MIMIC and dashed squares denote the private cohort.

The figure complements the annualized QRS and ST-T summaries reported in the main text by showing how these differences emerge across the individual forecast states. Low-tier trajectories remained comparatively close to baseline, whereas Intermediate- and High-tier trajectories showed progressively larger regional deviations with increasing forecast horizon. The overall pattern indicates that the tier ordering was not driven by a single forecast state but was expressed across the generated trajectory.

Endpoint-level cross-sectional operating metrics

Endpoint-level AUROC remained comparatively stable across forecast states, consistent with the high macro AUROC retention reported in the main text. In contrast, fixed-threshold operating characteristics changed systematically with forecast horizon, with sensitivity tending to increase and specificity tending to decrease.

The distinction between discrimination and operating-point behavior is important. Stable AUROC indicates that relative separation between endpoint-positive and endpoint-negative ECGs was largely retained across generated states, whereas the shift in sensitivity and specificity indicates that more generated waveforms moved toward the classifier-positive side of the fixed decision threshold with increasing forecast horizon. The endpoint-level results show that this pattern was not confined to a single dataset or endpoint.



Supplementary Fig. 4 Endpoint-level AUROC and dataset-specific macro sensitivity, specificity, and F1 across baseline and forecast states in MIMIC, CPSC2018, and Georgia. The main text reports the corresponding macro retention summary.